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Stakeholder subjectivities regarding barriers and drivers to the introduction of utility-scale solar photovoltaic power in Brazil

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ABSTRACT

Barriers and drivers of renewable energy systems are contingent upon particular technologies, social organizations, and institutions. Identification of barriers and drivers is necessary to devise policies and strategies to facilitate the introduction and dissemination of renewable energy generation. In 2002 the Brazilian government encouraged the adoption of renewable energy technologies such as wind, biomass and small hydroelectricity. Beginning in 2014, Brazil's federal regulatory agency started auctions aiming to develop ~12 GWp of utility-scale solar photovoltaic (PV). This paper explores the barriers and drivers to introduction of the solar photovoltaic technology in Brazil by focusing on the analysis of empirically determined subjectivities among the electricity power sector actors in the case of the Minas Gerais state. We identify and describe three perspectives (factors) using Q-method: (1) "We can do it" (2) "Step-by-step"; and (3) "It's not the money." Within these perspectives, planning for siting power plants, lack of transmission network, and biodiversity impacts were identified as three main statistically significant and highly ranked barriers. Identification of social perspectives may avoid conflict barriers to introducing utility-scale PV and suggest socially acceptable solutions for technical and economic issues.

1. Introduction

Implementing renewable energy technologies (RETs) is a policy priority in many countries, but governments face many complex barriers in making RETs viable technically and efficient economically. The gap between potential and actual RET conversion is partly the result of technical and economic barriers to deployment resources (Eleftheriadis and Anagnostopoulou, 2015; Reddy and Painuly, 2004). One optimistic view is that only "social and political" barriers impede the move toward complete global reliance on wind, water, and solar power by 2050 (Jacobson and Delucchi, 2011).

Several factors are responsible to creating barriers to RET expansion. One group of scholars has indicated the need for a technical-institutional paradigm change in the traditional power sector because RET demands structural, social, organizational, and economic changes (Wolsink, 2013, 2012; Tsoutsos and Stamboulis, 2005). For example, Wustenhagen et al. (2007), p. 2685) argue that power-generating firms "influence public policies and the ability of firms seeking to enter the power sector to access distribution infrastructure." Wolsink (2012) reports strong resistance to change firm behavior regarding penetration in the electricity sector. Del Rio and Unruh (2007, p. 1499) attest that "pre-existing infrastructure, both physical and institutional, can create important constraints on the adoption patterns of new technologies." For Brazilian utility-scale photovoltaic (PV), scientists report that low auction prices and weak government subsidies are barriers (Corrêa da Silva et al., 2016; de Jong et al., 2015; Souza and Cavalcante, 2016). Other reports indicate strong future growth in Brazilian solar (IEA, 2016) and highlight challenges regarding transmission capacity (REN21, 2015)

The challenge of reducing technical and economic barriers has attracted the attention of entrepreneurs, scientists and politicians around the world (Pinto et al., 2016; Echegaray, 2014; Matos and Silvestre, 2013). However, policy makers tend to offer solutions that do not apply to attending environmental, social, organizational, and political problems (Pereira et al., 2013; Zoellner et al., 2008). As a consequence, introduction of RETs is delayed in favor of maintaining traditional energy conversion technologies (Wolsink, 2013, 2012). Detailed empirical findings about social perspectives among power-sector stakeholders offer the opportunity for policy recommendations aimed at overcoming barriers and maximizing PV dissemination (Sindhu et al., 2016).

Here we determine subjectivities in the Brazilian power sector

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among entrepreneurs and officials in power regulation, generation, distribution and transmission regarding barriers and drivers for utility-scale PV. Identifying subjectivities or social perspectives may improve knowledge of barriers to photovoltaic introduction and help identify specific rationales influencing stakeholders and their ability to articulate favorable attitudes toward PV introduction and diffusion in Brazil.

2. Background

2.1. Barriers to RET expansion

The literature identifies several specific barriers to RET expansion. The existing electricity distribution network is cited as a barrier by several authors, who emphasize conflicts of interest and market power as slowing RETs (Wolsink, 2013, 2012; Jacobsson and Bergek, 2004; Unruh, 2002). For example, Wolsink (2012), p. 1811) argues that pressure exerted by incumbent firms "try to use their influence in the crucial political decisions" about system design, grid access, and RET incentives.

Workforce and information are also key barriers to RETs (del Río and Unruh, 2007). Unruh (2000) emphasizes the need to train technicians and professionals capable of supporting technological introduction and diffusion. Solar insolation maps are a key aspect of information availability needed to attract PV firms. As Martins and Pereira (2011), p. 4388) argue, "global investors may not know about the existence of viable solar sites.".

Incentives and cost barriers are cited by many authors, such as Martins and Pereira (2011), p. 4388), who argue that RET "incentive programs must be devised at municipal, state and federal levels." For these authors, defining a reference value for the price of solar and wind energies allied with tax reductions and exemptions on equipment and profits of companies operating on these renewable technologies were especially important. Persistence of negative incentives, such as subsidies directed to traditional power technologies, is another problem that may delay innovations (Zhai and Williams, 2012; Jacobsson and Bergek, 2004). Many authors agree on the need for governments to offer better purchase prices for solar power (Martins and Pereira, 2011; Goldemberg and Coelho, 2002) and other subsidies to offset the relatively high price of PV power (Corrêa da Silva et al., 2016; de Jong et al., 2015; Souza and Cavalcante, 2016).

Planning and siting procedures "influence payback time or delay government subsidies that may be available to investors" (Frate, 2015, p. 2), but Wustenhagen et al. (2007), p. 236) note that "traditional power-generating sectors may not have the institutional culture of planning and siting", which is a barrier to implementing RET. Montezano (2012) argues that regional scale studies for siting PV are important because of the relatively small area of proposed RET projects and relatively high cost of individual siting studies.

Land requirements, which are relatively high owing to low power densities of PV (Smil, 2015), represent another barrier. Large land purchases may be expensive (Shah et al., 2015). Attachment to place may help generate conflicts over land; Wolsink (2012) noted that land is not only the site of the conflict, but also the origin of the conflict. In coastal Brazil, land-tenure insecurity is a key reason for wind power conflicts (Brannstrom et al., 2017). Concerns over biodiversity impacts are closely related to the land issue, as demonstrated by conflicts in California solar power development (Storms et al., 2013). Biodiversity impacts include landscape fragmentation (Nunes and Meyer, 2014; Tsoutsos et al., 2005) and soil disturbance (Hernandez et al., 2014). Fthenakis and Zweibel (2003) show that PV power plants require water at volumes of 0.02 m^3/MWh for cleaning.

2.2. Brazilian RET policies

Brazilian electricity generation is composed mainly of hydroelectricity (64%), thermal (natural gas, coal, biomass and nuclear; 33%), while solar PV accounts for 0.2% (MME, 2016). High reliance on hydropower generation helped produce social movements opposed to social and environmental impacts of large dams in the Amazon region (Pereira et al., 2013; Farias, 2014; Rothman, 2001; McCormick, 2007); in addition, severe drought in 2001 sharply increased electricity prices, creating major political difficulties and putting economic development targets beyond reach. Pinto et al. (2016) attest that throughout the last 20 years, the heavy reliance on hydropower has not been seriously questioned, but since 2002 the Brazilian government has expanded and diversified its energy portfolio focusing on techno-institutional developments, research-development projects and public policies. Juarez et al. (2014), p. 833), describe the insertion of RET into Brazil's grid as "a win-win situation for society, energy firms and the environment" while Ribeiro et al. (2016), p. 554) point out that RET "is viewed by society as a positive alternative to support economic and social development". Goldemberg and Coelho (2002), p. 55) even argue that RETs can "help Brazil reduce poverty and inequality."

Utility-scale PV in Brazil is comprised of 44 plants in operation (28 MW potential) and 21 plants under construction (616 MW), with 90 plants (2.4 GW) in planning stages (ANEEL, 2017). Observers predict that PV will play an important role in the future, from 0.5 GW (0.2%) in 2017–3.5 GW (1.8%) in 2023, likely to exceed biomass and small hydro power (Empresa de Pesquisa Energética (EPE), 2014a). Growth of PV was stimulated by Brazil's Ministry of Energy and Mines (MME), which held an auction in 2014 dedicated to the purchase of energy produced by PV. This auction received bids of 10.79 GW (400 projects), of which 0.89 MW were contracted (3 projects) at mean price of R$ 215 per MWh (~US$72 in early 2015 and ~US$54 in early 2016), nearly 18% less than the opening price. The investment for 0.89 GW is approximately R$ 4.14 billion (~US$1.38 billion and US$1 billion in early 2015 and early 2016, respectively) to R$ 4.7 million per MW (~US $1.57 billion and US$1.18 billion in early 2015 and early 2016, respectively). A second auction took place 14 August 2015 with the expectation that 1.27 GW (36 utility-scale projects) will be contracted for Minas Gerais state (Empresa de Pesquisa Energética (EPE), 2014b).

PV has high public acceptance in Brazil because it is associated with a permanent energy source and is not known to cause negative socio-environmental impacts. This idea is produced mainly by government and international development agencies, universities and power plant entrepreneurs (Carneiro, 2000; Locatelli, 2011; Farias, 2014), which seek Brazilian media outlets as means to legitimize their sustainability discourses to create and expand their participation in energy markets. As a consequence the media portray the idea that rapid diffusion of RETs is essential and desirable in comparison to hydroelectric power, which has created strong social opposition (Rothman, 2001; McCormick, 2007; Locatelli, 2011) while generating less power because of low reservoir levels.

Governance of Brazil's electricity sector is dominated by state-run firms and private firms that bid on electricity supply auctions. These groups are relatively insulated politically from universities and social organizations. Identification of different views or social perspectives among representatives of the electricity sector may create a more holistic understanding of barriers and drivers regarding a specific RET, which is seen almost exclusively in highly positive and non-problematic terms.

However, the literature on Brazilian solar power does not yet describe stakeholder perceptions and discourses about utility-scale PV. This is an important gap in knowledge because the addition of new hydropower capacity is highly problematic on grounds of efficiency and environmental costs (Corrêa da Silva et al., 2016), while new coal and nuclear power plants are not justified economically or environmentally (de Jong et al., 2015). Studies aimed to discuss social perspectives in the Brazilian's electricity sector are necessary to determine empirically diverse stakeholders' views on barriers to maximize drivers to PV dissemination and diffusion (del Río and Unruh, 2007). Scholars have indicated that "government incentives" and a "favorable business

environment" are necessary for increased solar power (Corrêa da Silva et al., 2016, p. 338), but the specific content of these claims, and the existence of other possible claims, are unknown.

3. Materials and methods

3.1. Study region

We carried out this social perspective study in Northern Minas Gerais state (NMG), Brazil, a region of high potential for solar photovoltaic power generation. NMG is predicted to become the third Brazil's large center of PV installed capacity. Brazil's National Agency for Electrical Energy, a federal regulatory agency that establishes rules for RET generation, transmission and distribution (ANEEL; Pinto et al., 2016), is running auctions to develop 12 GWp of solar photovoltaic (PV) utility-scale photovoltaic in Brazil. The selected firms that comply with the installation schedules will sell their power to Centrais Elétricas Brasileiras (Eletrobrás) under a thirty-year feed-in-tariff contract. The NMG presents the highest state levels of solar radiation and will deploy ~1 GWp power plant capacity (CEMIG, 2012). NMG has a total area of 72,440 km^2 and population of about 1.1 million inhabitants (IBGE, 2013). Montes Claros, the region's main urban center is location for some medium to large industries, while across the region there are dozens of higher and technical education institutions as well as various farmers' associations, civil society organizations, and local and regional development agencies (Gonzaga, 2014; Oliveira, 2013; Drummond, 2011; Diniz et al., 2011).

Northern Minas Gerais is mostly comprised of large flat expanses of degraded pasture with the state's highest solar radiation levels from 4.97 to 5.26 kWh/m^2/day (CEMIG, 2012). Maximum annual average temperature is 32 °C and minimum annual average temperature about 27 °C. Average relative humidity is 60%. This region is part of Brazil's semi-arid climate regions, in the transition zone between the *Cerrado* (woody savanna) and *Caatinga* (dry shrubland) biomes. With 89 municipalities (roughly equivalent to U.S. counties), NMG covers 139,000 square kilometers and has per capita GDP of BRL7260, about half the average of Minas Gerais state. The HDI for the region is 0.529, far below the 0.731 mean for Minas Gerais State. NMG received national prominence in the 1970s when Brazil's first massive agricultural irrigation project aimed to produce food based on 700 km of water channels for distributing irrigation water (Florin et al., 2013).

3.2. Study design: Q-method

We employed Q-method, which combines qualitative and quantitative techniques, to describe and analyze subjectivity among key actors in the electricity sector regarding their views on barriers to photovoltaic introduction in Brazil's Northern Minas Gerais region. Q-method relies on four main steps well described in the literature (Robbins and Krueger, 2000; Eden et al., 2005). First, we created a concourse of statements from 18 semi-structured interviews regarding utility-scale PV with two representatives of local entrepreneurs, one representative of a farmer group from NMG, two representatives of a local social movement, two scholars based in the region, and eleven representatives of municipal and state governments. Interviews were carried out between 11 and 19 November 2014 in Belo Horizonte (MG), Montes Claros (MG) and Jaíba municipality (MG), which is planned to host 360 MWp PV utility scale facility. Interviews lasted between 30 and 60 min each. The guiding question for interviews was, "Which are the main barriers and drivers to utility-scale solar photovoltaic technology introduction in the northern region of Minas Gerais?" Interview transcripts were used to identify 103 statements on the basis of balance among respondents and wide spectrum of views. Five social scientists knowledgeable about solar energy technology issues, energy and the environmental policies of the region, scored these statements regarding suitability for this study. They relied on criteria of relevance, frequency, and similarities. Eventually, this scoring process arrived at 16 statements for the Q-set (Table 1).

Table 1
Q-sort distribution for 16 statements.

	Most disagree			Neutral		Most Agree	
Value	−3	−2	−1	0	1	2	3
Frequency	1	2	3	4	3	2	1

We then moved to the second phase, in which we identified potential respondents among people who had signed up for the 2015 International School on Solar Energy (ISSE). The ISSE was promoted by University of Brasília (UnB), the Jaíba Project for Research and Development (funded by ANEEL and industry contributions), and the German Cooperation for Sustainable Development, with partnerships including Brazilian and international universities, power companies, international agencies, industries, government and international embassy services. The 2015 ISSE aimed to explore the technological issues of solar energy conversion through photovoltaic and thermal systems, including also the availability of solar resources at ground level. The 2015 ISSE in Brasília attracted professionals, engineers, graduate and undergraduate students in different fields of solar energy to participate in lectures, round tables and hands-on workshops activities (http://www.ses-unb.net).

We selected 16 respondents from high level professionals in Brazil's electricity sector, including producers, distributers, transmitters and regulators to sort the statements into a quasi-normal distribution during 23–26 February 2015. All participants were people who had first-hand knowledge on utility-scale solar PV and were knowledgeable about technical, sociopolitical, ecological and institutional environments of NMG. Participants were also selected based on whether they were willing to participate and their level of involvement with RET research, development and innovation. Respondents ranked the statements in a semi-normal distribution that forced sorters to place the statements into a grid that allowed for only one statement corresponding to "most agree with my views" (+3) and only one for the "most disagree with my views" (−3), while four statements were permitted in the "neutral" category (Table 1). The first author was present during the sorting, and discussed the procedure with each respondent, who was encouraged to rank the statements according to his opinion. Each sort lasted approximately 40 min.

To aid in the interpretation of factors, we conducted post-sort interviews with each respondent to elicit the rationale for statement rankings. These interviews were recorded and lasted approximately 25 min. We transcribed interviews and inserted them in Atlas TI software, which assisted in the coding process so that we could identify repeated words among respondents.

The third phase of Q-method involved the correlation and factor analysis using dedicated freeware (Schmolck, 2002) to determine the correlation matrix, extract and rotate (through VARIMAX rotation built into the freeware) significant factors, and calculate statement z-scores for each factor. We selected a three factor solution that accounted for 51% of variance (Tables 2 and 3). We examined a four-factor solution, but it explained little additional variance and increased error; a two-factor solution explained too little variance and provided excessively simplistic factors.

The fourth step of Q-method is to interpret factors, sometimes accompanied by a validation phase with "loaders" on factors (Table 4). We refer to Statements by number and respondents by number in the section below. In Q-method, distinguishing statements represent statistically significant differences between the rank in one factor compared to all other factors at $P < 0.01$ or $P < 0.05$. We consider factors to be "stakeholder perspectives."

Table 2
Factor characteristics.

	Factor 1	Factor 2	Factor 3
Eigenvalue	3.397	2.585	2.186
No. of defining variables (sorts)	5	4	4
% variance explained	19	17	15
Average relative coefficient	0.800	0.800	0.800
Composite reliability	0.952	0.941	0.941
Standard error of factor scores	0.218	0.243	0.243
Distinguishing statements	1, 3, 7, 11, 15	2, 5, 6, 7, 10, 11, 13, 14, 15	4, 7, 9, 11, 12, 15

Table 3
Correlations between factor scores with 98% confidence interval.

Factor	1	2	3
1	1.0	0.1023 ± 0.2474	0.0871 ± 0.2481
2	0.1023 ± 0.2474	1.0	−0.076 ± 0.2486
3	0.0871 ± 0.2481	−0.076 ± 0.2486	1.0

Table 4
Rotated factor loadings (* indicates defining sort) with respondent type summarized.

Sort	Type	Factor 1	Factor 2	Factor 3
1	Regulator	−0.2218	0.2992	−0.0989
2	Entrepreneur	0.5552*	−0.0611	0.0726
3	Entrepreneur	0.4873	−0.6481*	0.2572
4	Regulator	0.2777	−0.3359	−0.0785
5	Power distribution	0.7986*	−0.0373	0.2402
6	Entrepreneur	0.0618	−0.0852	0.5218*
7	Regulator	0.0875	0.0175	0.7306*
8	Power production	−0.1156	−0.6356*	−0.3604
9	Power transmission	0.5856*	0.0923	−0.1853
10	Entrepreneur	0.5274*	0.1838	0.3684
11	Regulator	0.552	0.4888	0.4665
12	Regulator	0.6612*	−0.0873	−0.3964
13	Power production	0.1284	0.8436*	−0.2593
14	Power production	0.4573	0.7567*	0.0319
15	Power distribution	−0.2026	0.3032	0.5467*
16	Entrepreneur	0.0172	−0.206	0.6814*

4. Results and discussion

We identified three social perspectives (factors) relating to potential barriers to PV development in Brazil: (1) We can do it; (2) Step-by-step; and (3) It's not the money (Table 5). These social perspectives are comprised of statistically significant defining statements that received statistically significant scores by respondents. In addition, the social perspectives are informed by respondents who "loaded" on particular social perspectives and the rationale they provided for their sorting of the statements. Below we describe these results in detail.

4.1. "We can do it if the price is right" (Factor 1)

The "we can do it" social perspective is defined by respondents representing regulatory agencies and electricity service providers and by statistical metrics of statements relating to the importance of auction prices for solar power as the most important driver for solar energy in NMG. This social perspective agrees strongly with a statement alleging that the low price level offered by auctions for solar power is a major obstacle (Statement 15; $P < 0.01$; $Z = 2.06$), a claim also made in the literature (Corrêa da Silva et al., 2016; de Jong et al., 2015). Among various justifications for agreement with this statement is the argument made by one regulator (respondent 9; Loading = 0.5856), who affirmed that "technical issues will be resolved when the economic and environmental issues are resolved." Another respondent (5; Loading = 0.7986) put the issue simply: "Unfortunately, everything depends upon financial motivation."

Other potential barriers received modest agreement, such as the idea that there is enough available titled land with low costs for PV diffusion in NMG (Statement 1; $P < 0.01$; $Z = 0.87$). For example, respondents noted that "land at a low price is a huge differentiating factor" (2; Loading = 0.5552) and PV's large land demands mean that "the low price of land is relevant" (5; Loading = 0.7986). Further support for putting auction prices first comes from the negative response to the claim that the highway system presents problems to introduce and diffuse technology in NMG (Statement 3; $P < 0.01$; $Z = -1.92$). Respondents argued that "equipment necessary for this technology may be transported easily with the existing highway system [in NMG]" (11; Loading = 0.5520).

Interestingly, this perspective gives low importance to socio-environmental sustainability (7; $P < 0.05$; Loading = −0.72). Respondents trusted that investors would "follow the [environmental] law," which presumably will assure sustainability (2; Loading = 0.5552) and suggested that in a trade-off between energy and conservation, energy would win: "It's obvious that they [conservation units] are important, but I don't think they are a bottleneck [because] we will have to expand and diversify our electricity supply" (12; Loading = 0.6612).

4.2. "Step-by-step" (Factor 2)

The "Step-By-Step" social perspective is characterized by statistically significant statements regarding potential obstacles that diverge from reports in the literature. For example, a key statistically significant and highly ranked statement defining this social perspective is agreement with the claim that the electricity transmission network lacks capacity to distribute the solar electricity that PV may generate (Statement 11; $P < 0.01$; $Z = 1.73$). Two electricity company respondents loading on this factor justified their response that the electricity network lacks capacity. One (14; F2 loading = 0.7567) argued that "solar plants established in places with high solar radiation [NMG] don't matter at all if there is no way to distribute energy." A second respondent (13; F2 loading = 0.8436) used more specific terms, arguing that "You can have good prices but if you don't have any way to distribute the electricity generated then there won't be investment in utility-scale solar plants." For this social perspective, transmission bottlenecks are more important to resolve than auction price, which was rated far lower (Statement 15; $P < 0.05$; $Z = 0.20$).

Another statement that characterized this social perspective was strong disagreement with the idea that high solar radiation in NMG is a sufficient condition for PV viability (Statement 2; $P < 0.01$; $Z = -1.34$). An electricity company representative (respondent 14) said that "solar potential is not the only sufficient and necessary condition to [solar] technology development." Similarly, a regulator (respondent 11; F2 loading = 0.4888) said: "[the high solar irradiation] is a necessary but may not be a sufficient condition in case, for example, you don't have government incentives for technology or if you have social rejection because of competition for resources such as water." Another regulator (1; F2 loading = 0.2992) pointed out that "[solar radiation] is a basic input into electricity generation but it doesn't help at all if there is no labor, if you have irreversible environmental impacts, if you don't bring benefits to the communities, if you don't have a means to distribute power."

Respondents also emphasized the need for a trained workforce. Strong agreement exists with the idea that lack of specialized labor harms the development of solar plants (Statement 5; $P < 0.01$; $Z = 1.49$). A regulator (11; F2 loading = 0.4888) agreed with this statement because "you need specialized technical capacity from the origin of the project until installation, operation, and maintenance." An electricity company representative (14; F2 loading = 0.7567) gave the highest

Table 5
Factor array and rank statement scores for Q-sort statements. Statistical significance at P < 0.05 indicated by *, while **identifies significance at P < 0.01. Non-significant "consensus" statements are indicated by underlined z-scores.

Statement	F1: We can do it		F2: Step by step		F3: It's not the money	
	Z	rank	Z	rank	Z	rank
1: Availability of titled land at low price is sufficient for PV expansion.	0.87**	2	−0.93*	12	−0.50*	11
2: High solar radiation in NMG is a sufficient condition for PV viability.	0.57*	6	−1.34**	16	0.57*	4
3: The highway system is a problem for PV development.	−1.92**	16	−0.58*	11	−0.21*	10
4: The lack of information regarding the possibility for socio-economic development for communities could harm PV development.	0.28*	8	0.32*	6	2.53**	1
5: Non-existence of specialized labor, with Bachelor's or high school degrees, specialized in solar energy, harms the development of solar plants.	0.58*	5	1.49**	2	0.09*	8
6: Overlaps of federal, state, and regional subsidies in the semi-arid region is necessary for sustainability of solar technology in NMG.	0.79*	3	−0.20**	10	0.99*	2
7: Low importance given to socio-environmental sustainability by developers of solar plants is an obstacle for diffusion of PV technology.	−0.72*	13	0.11*	8	0.99*	3
8: Requirements for technology transfer by Brazilian law is an obstacle to consolidate the solar production chain.	−0.60	12	−1.16	15	−1.18	15
9: Lack of knowledge among international investors about solar potential in NMG reduces the possibility of technology diffusion.	−1.58*	15	−1.15*	14	0.05**	9
10: Studying alternative locations for solar plants by cartographic and geoprocessing means is a condition for sustainable solar development.	−0.19*	10	1.41**	3	0.16*	7
11: The electricity transmission network lacks capacity to distribute solar electricity.	−0.24*	11	1.73**	1	−0.91*	13
12: The presence of conservation units and their buffer zones are an obstacle for solar expansion.	0.41*	7	0.65*	4	−0.63**	12
13: Loss of habitat for arachnids, reptiles, mammals, and birds affects the sustainability of solar electricity plants.	0.69*	4	−1.07**	13	0.28*	5
14: Water shortage in NMG is a large incentive for solar electricity development.	−0.90*	4	0.57**	5	−1.54*	16
15: The low price offered by Minas Gerais auctions for solar electricity is the greatest obstacle.	2.06**	1	0.20*	7	−0.93*	14
16: Investment in specific links of the solar electricity productive chain represents important support for sustainability	−0.09	9	−0.05	9	0.23	6

rating to this statement, warning that "lack of specialized labor will result in operational faults, which could support more consolidated technologies such as hydroelectricity." A regulator (1; F2 loading = 0.2992) agreed with the statement, but for reasons relating to local socio-economic relations: "the production of electricity will attract trained workers and the firm could train the workforce, but it is more attractive to train people from the region [NMG] than bring in workers from outside." Another regulator (4) put the socio-economic issue more bluntly: "It's important that local people become friends of the plant because in these more remote sites you can't get the workforce to stay."

A fourth statistically significant statement refers to the need to identify locations for solar plants using cartographic and geoprocessing for sustainability (Statement 10; P < 0.01; Z = 1.41). Strong agreement with this technical approach to siting PV plants is justified by a regulator (respondent 1; F2 loading = 0.2992), who justified a high rating by arguing that "Any project needs to minimize impact through evaluation of several different sites." An electricity company representative (respondent 14; F2 loading = 0.7567) noted similarly that "this is a necessary condition for any project. Without this information there is no way to proceed." Another regulator (11; F2 loading = 0.4888) replied, "having detailed information is fundamental to avoid local socio-economic or technical problems."

Finally, this social perspective disagreed with the statement that the loss of habitat for arachnids, reptiles, mammals, and birds affects the sustainability of solar electricity plants (Statement 13; P < 0.01; Z = −1.07). One electricity company respondent did not believe that solar PV technology would affect these types of animals (13; F2 loading = 0.8436). Respondents who loaded on F1, "We can do it," also articulated reasons for a negative rating to this statement. For one respondent, the trade-offs between energy and biodiversity were real, there were sound reasons for siding on energy: "in spite of the importance of biodiversity, one has to consider [Brazil's] energy deficit" (5; F1 loading = 0.7986). Another respondent suggested ways to overcome the potential loss of biodiversity, indicating that "there are always ways to do a study, collect species, and minimize impacts, and we have to remember that we need to produce energy one way or another."

4.3. "It's not the money" (Factor 3)

"It's not the money" is defined primarily by concern for poor dissemination of information about advantages and disadvantages of solar power as a barrier to PV, and secondarily by concern that firms ignore local socio-environmental concerns. For example, statement 4, which posited that the "lack of public information available to communities could harm the development of solar plants" (P < 0.01; Z = 2.53), was ranked very highly. One entrepreneur phrased this idea eloquently, arguing that "It is important for the sustainability of solar technology to generate consciousness in the communities so that [community residents] know the potential benefits that solar resources may offer" (6; F3 loading = 0.5218). In contrast with "We can do it if the price is right," responses were negative to the statement alleging that the low auction price is a major obstacle (15; P < 0.05; Z = −0.93). As one entrepreneur (6; Loading = 0.5218) noted with a −3 ranking to statement 15, "the auction price is a result of good sectorial planning."

The social and community aspects were striking for "It's not the money," in contrast with reports of PV barriers in the literature (Corrêa da Silva et al., 2016; de Jong et al., 2015). A representative of a regulatory agency explained that "if the communities are not involved and not informed about the possible benefits that [solar power] technology could promote, [then] you will have a negative social initiative as we see around hydroelectric power plants" (respondent 11; loading = 0.4665). By making reference to Brazil's substantial anti-dam social movement (Rothman, 2001; McCormick, 2007), this respondent offered a warning to PV developers, which a representative of an electricity company (14) related: "If it is possible for the state or the firm to inform the public before public hearings, this could facilitate [community acceptance], otherwise not informing could generate negative reactions." Related to this concern is the relatively high ranking to the statistically significant statement, "The low importance given to socio-environmental sustainability by developers of solar plants is an obstacle for diffusion of PV technology" (7; P < 0.05; Z = 0.99).

Other statistically significant statements that characterize "It's not the money" relate to price and infrastructure, both of which require strong government involvement. One is a negative reaction to the claim

that the transmission system lacks capacity to handle energy that will be produced (Statement 11; $P < 0.05$; $Z = -0.91$). Respondents loading on this factor argued that "even though it [NMG] lacks capacity, this is a technical issue that could be resolved with government investment" (07; Loading = 0.7306). In more specific terms, a respondent argued that "If the diffusion of photovoltaic energy occurs, then the regulatory framework will easily permit contracting for the expansion of transmission lines" (11; Loading = 0.4665).

5. Conclusions and policy implications

We identified three empirically determined and statistically significant social perspectives among Brazilian energy sector representatives. The content of these three social perspectives does not map precisely onto views of RETs reproduced by Brazilian media. Our findings indicate that infrastructure, poor community outreach, and lack of planning are perceived as potential barriers to PV expansion in Brazil, while auction prices and government subsidies are viewed as more modest barriers than previously reported (Corrêa da Silva et al., 2016; de Jong et al., 2015; Souza and Cavalcante, 2016). These subjective views of drivers and barriers do not precisely mirror previous work on RETs in other developing countries (Sindhu et al., 2016).

For "It's not the money", concerns for socio-environmental issues relating to PV development and dissemination of information to communities are highly important. Experiences from Brazil's anti-dam protest movement offer a lesson that PV developers should learn if they want to avoid a major socio-political barrier. Notably, planning and siting procedures used for hydropower projects have been reproduced in RETs (Farias, 2014; Locatelli, 2011; Rothman, 2001; McCormick, 2007). "It's not the money" includes the concern that the utility-scale PV sector is ignoring community information, which could create dozens of similar problems to the ones that hydroelectric dams created. Indeed, decision makers have tended to ignore views of local stakeholders in siting decisions (Frate, 2015; Farias, 2014; Meireles et al., 2013; Locatelli, 2011). Martins and Pereira (2011), p. 8346) argue that "lack of community information is a barrier regarding for social acceptance of RETs in Brazil." Overall, this social perspective shows a new environmental consciousness among stakeholders in Brazil's energy sectors that go beyond the auction prices and technical infrastructure issues.

By contrast, "We can do it" sees low auction prices for solar energy as the major barrier to PV development, which mirrors reports in the literature. "We can do it," unlike "It's not the money," believes barriers are limited to government responsibilities, such as auction prices, price of titled land, highway system, and transmission capacity. For "We can do it," high auction prices are the most important driver of utility-scale PV in NMG; this is based on the view that financial motivation is more significant than technical and socio-environmental issues. A specific point of divergence is transmission capacity, which "We can do it" believes is easily contracted and built. Cheap titled land is enough to promote PV development.

"Step-by-step" is far more concerned about barriers to utility-scale PV expansion involving specialized labor and transmission capacity, similar to findings reported by Sindhu et al. (2016) for Indian PV. This group is concerned about the relationships between the utility-scale PV, technical-infra-structure, labor and nature. These are real concerns; for example, Almeida (2015), p. 86) shows that the "Jaíba Solar will produce 2.78 MW at peak even though the local transmission grid capacity is 1.25 MW". Other Brazilian analysts have noted the need for broad technical-institutional paradigm change in the traditional power sector to accommodate RETs (Pinto et al., 2016; Echegaray, 2014; Diniz et al., 2011).

Labor requirements indicated by "step-by-step" are present in the literature (Diniz et al., 2011; del Río and Unruh, 2007). Echegaray (2014), p. 131) predicts that a positive future because of "significant funds invested in labor specialized in PV installation in Brazil." Diniz et al. (2011) argue that in NMG there are dozens of higher and technical education institutions as well as various farmers' associations, civil society organizations, local and regional development agencies able to assist in workforce development. But a more fundamental issue is that this group's ignorance or dismissal of biodiversity impacts suggests the need to inform decision-makers in utility-scale PV about environmental justice and environmental law.

Our findings suggest that previous reports may have over-emphasized auction prices and government subsidies as barriers to utility-scale PV. A broader notion of stakeholders, coupled with a new instrument for data collection and analysis, has revealed new barriers. Some of these barriers, such as skilled labor, are common in other cases of PV development; but the framing of potential social conflict in hydropower terms is more unique to Brazil's energy experience.

In conclusion, these findings help advance knowledge on barriers to utility-scale PV development and illuminate knowledge domains unknown or poorly discussed among stakeholders in the traditional energy sector. Knowledge of social perspectives provides better understanding of the social capacity to introduce RETs and promotes better information about RETs to energy scientists, policy makers, investors, and nearby communities.

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